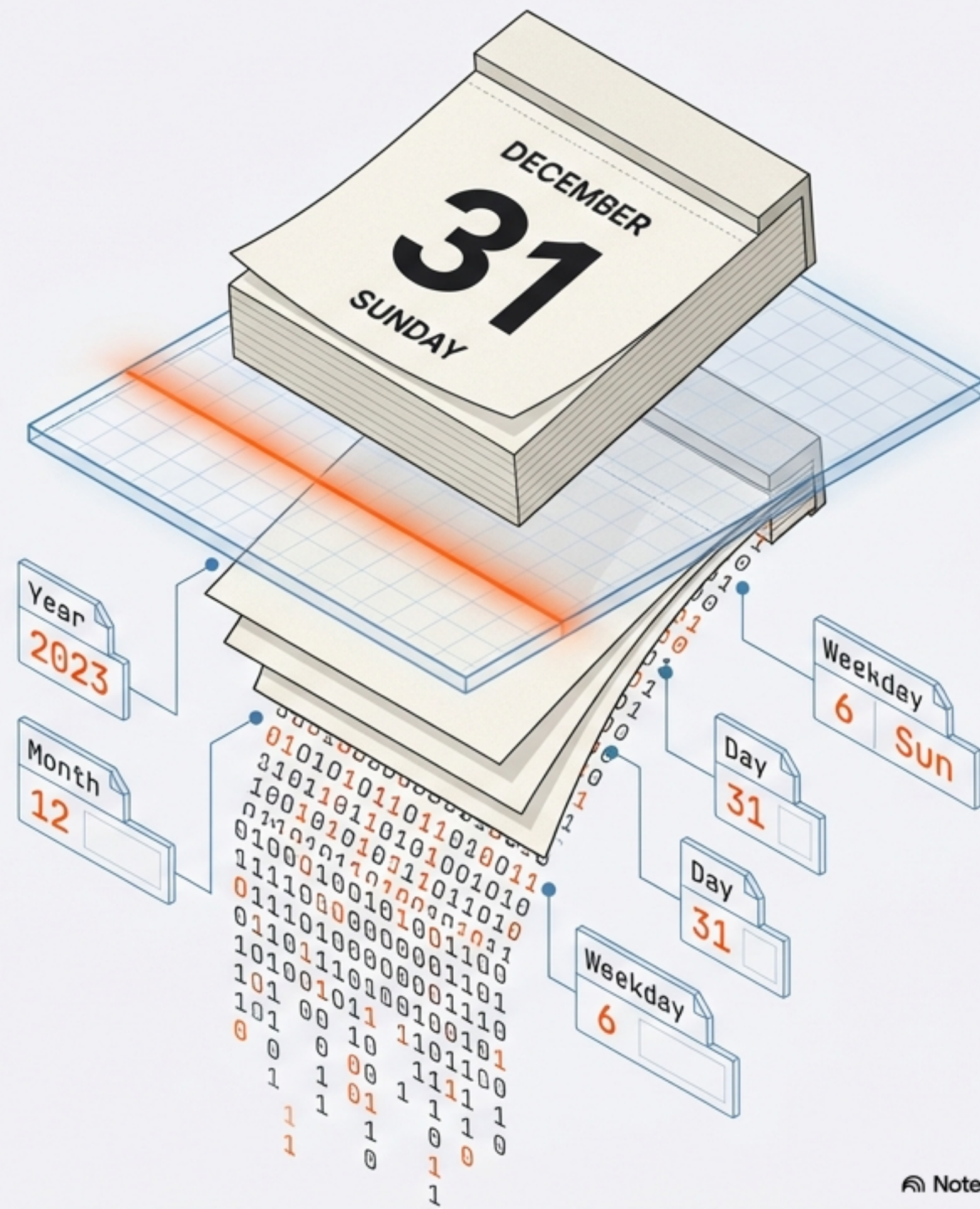


Feature Engineering: Handling Date & Time

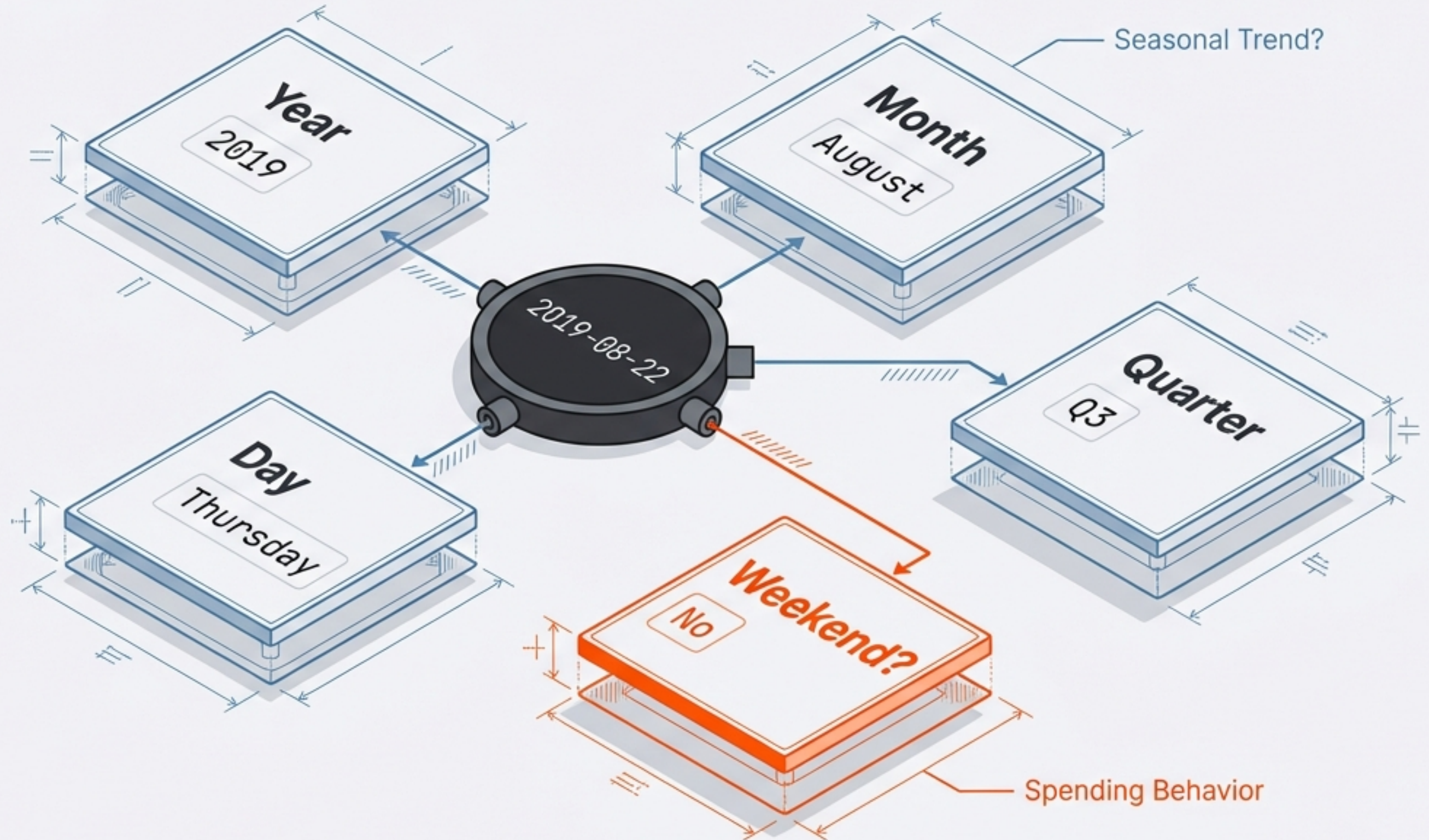
A technical guide to extracting hidden insights from temporal data in Python Pandas.

Source: CampusX Curriculum



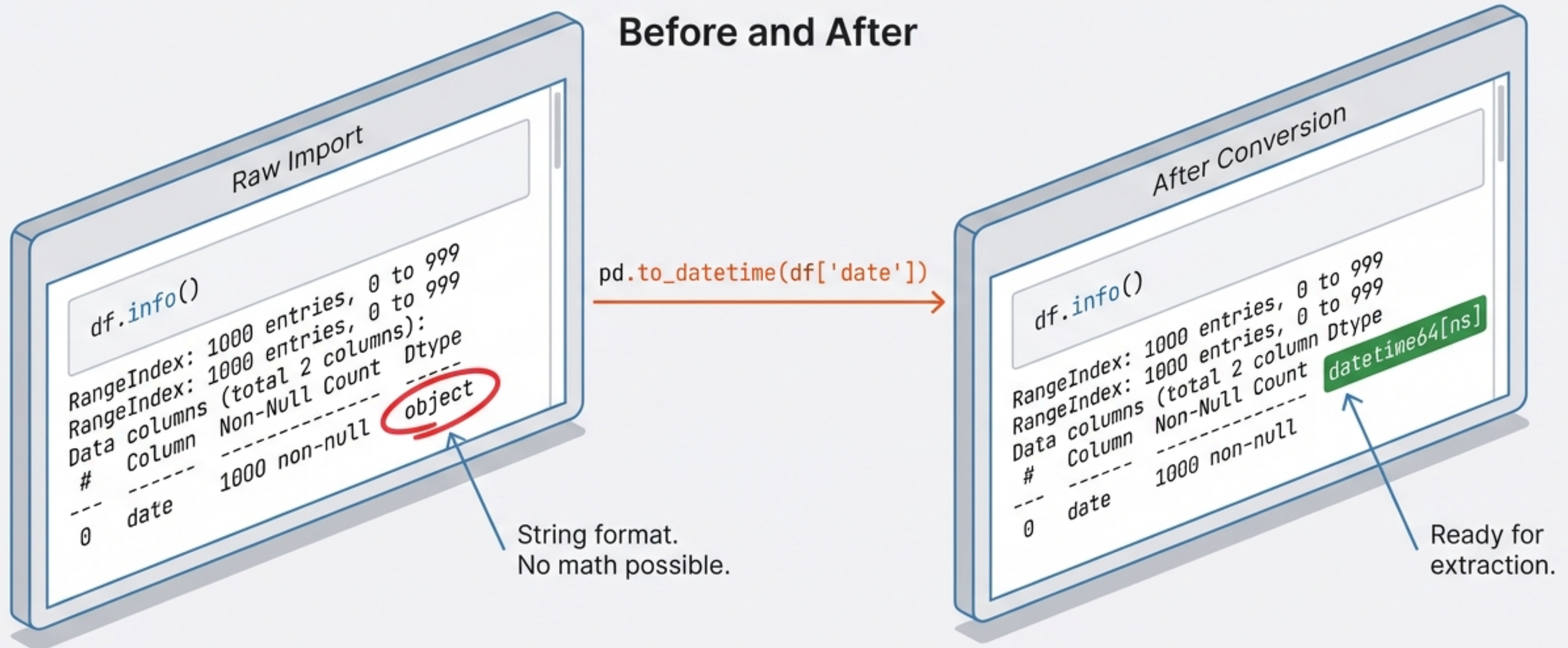
A Date is a Compressed File of Features

A single raw date string contains layers of hidden context vital for ML models. In an Expense Tracker app, knowing **when** a user spends is just as important as **how much**.



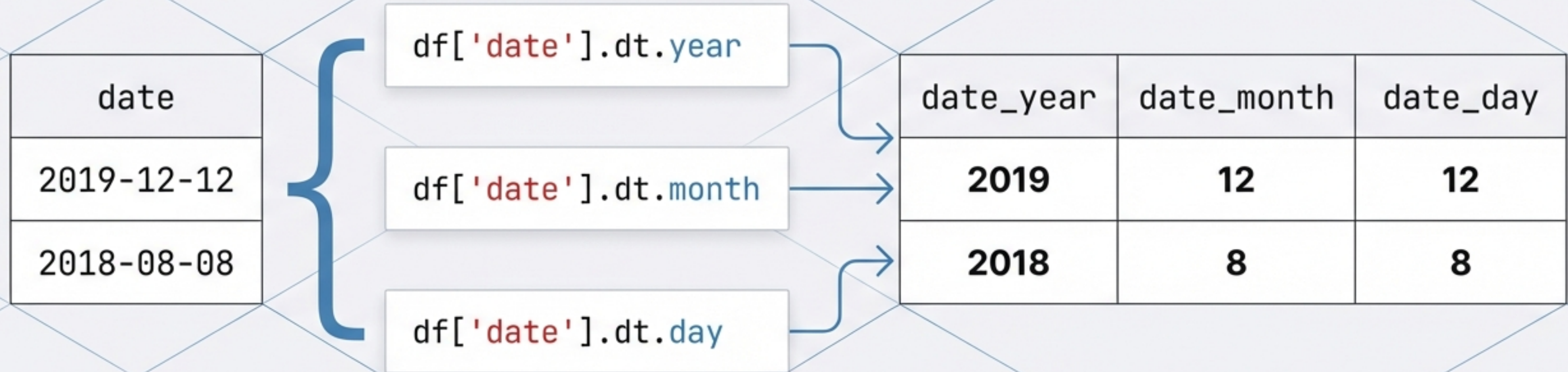
The Silent Error: Python Treats Imported Dates as Strings

Before extraction, data must be converted using `pd.to_datetime()`.



Extracting the Fundamental Units

The .dt accessor allows direct access to numerical components.

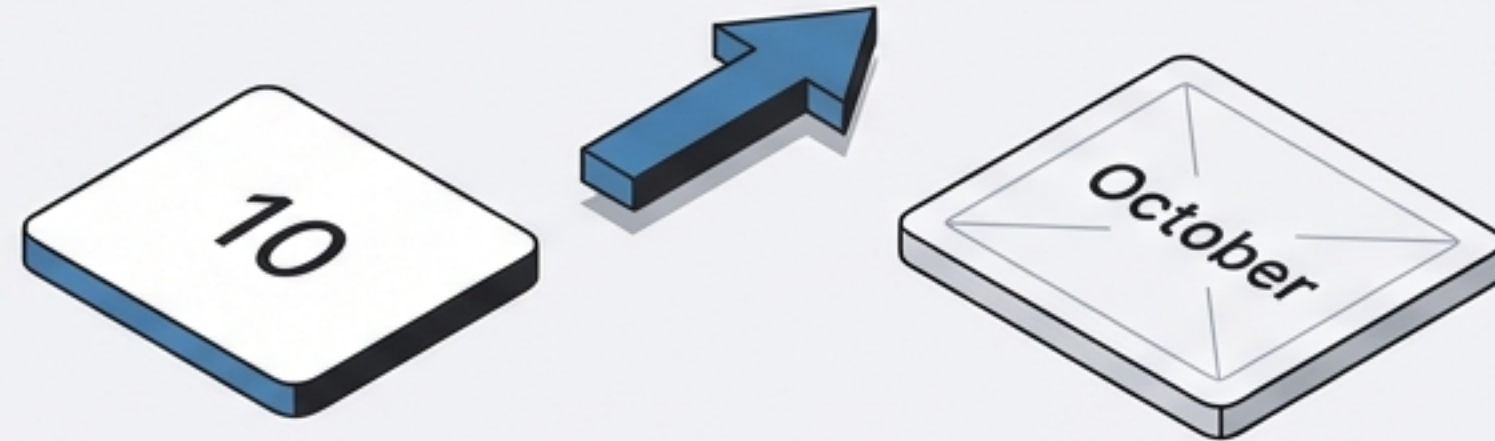


Converting Numbers to Names

Improving interpretability for categorical analysis

Month Name Extraction

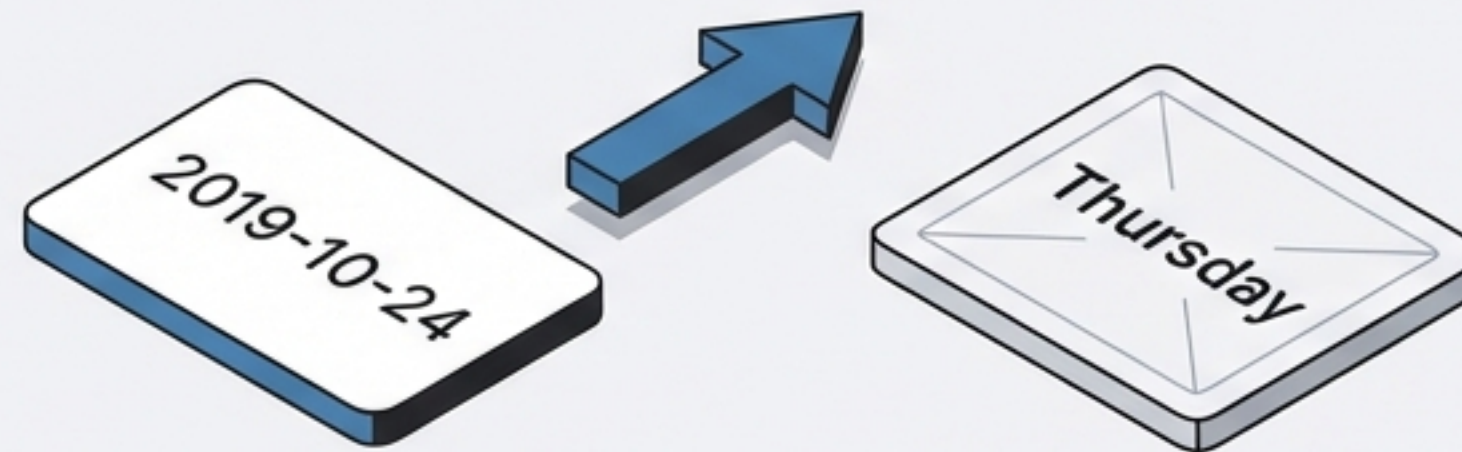
```
df['date'].dt.month_name()
```



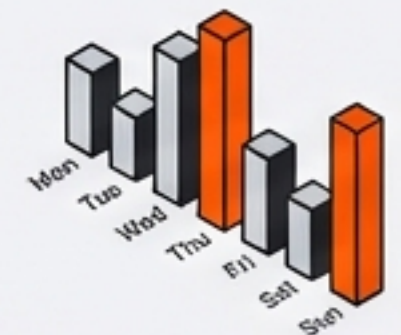
Useful for seasonality plots.

Day Name Extraction

```
df['date'].dt.day_name()
```



Vital for weekly cycle analysis.



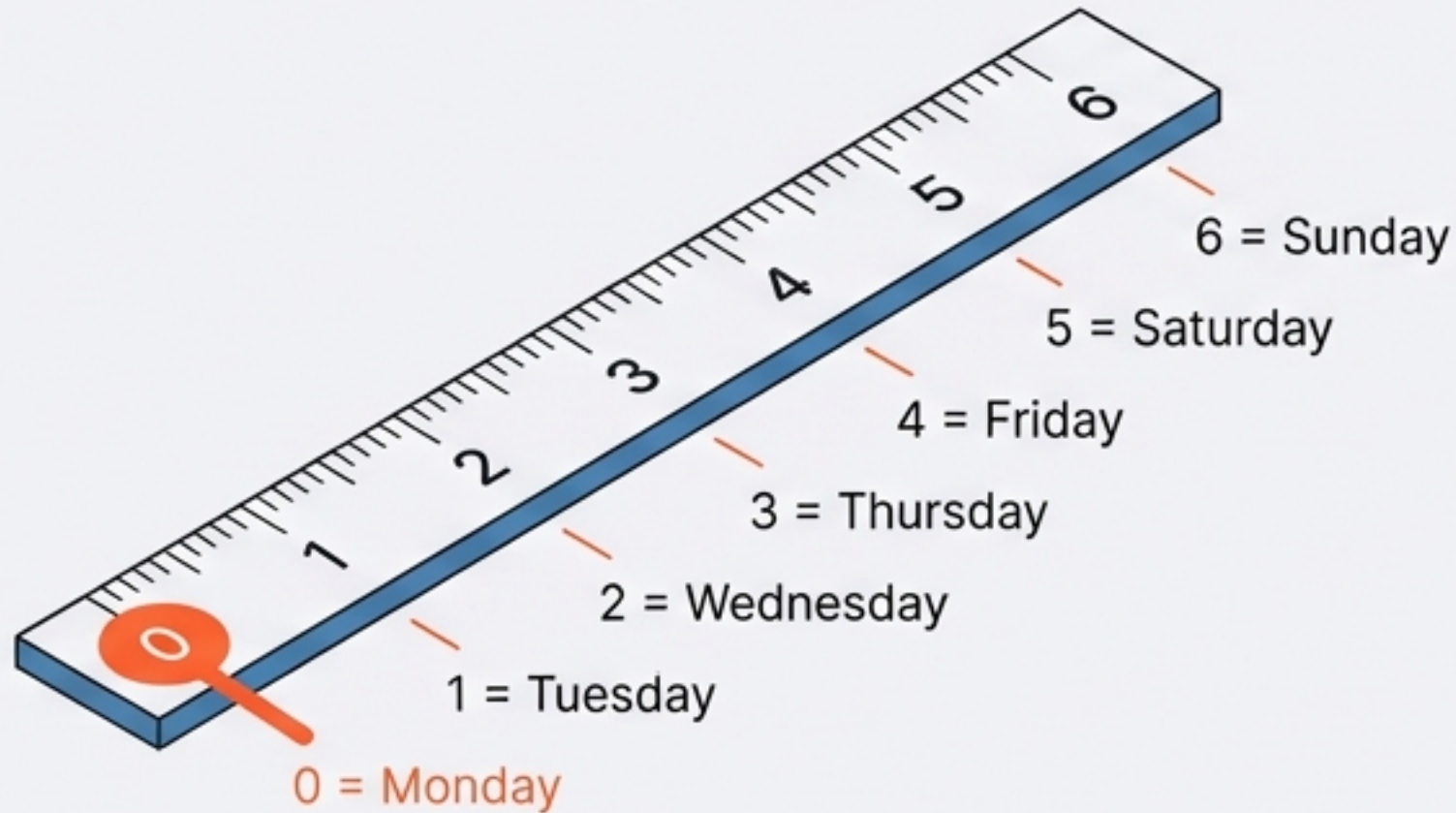
Sales by Day Name

Determining Calendar Position

Placing the data point within the structure of a week or year.

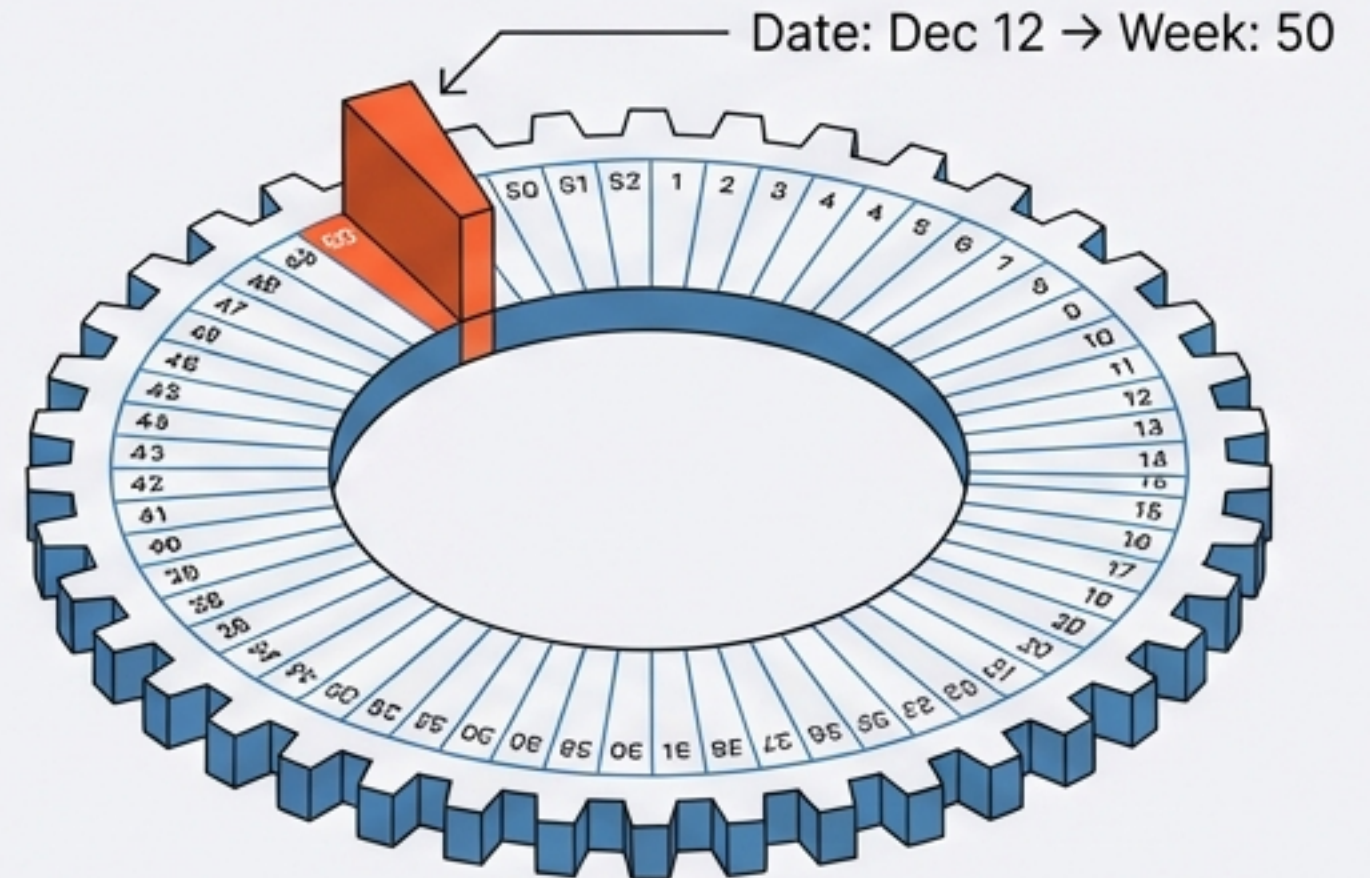
Day of Week Index

```
df['date'].dt.dayofweek
```

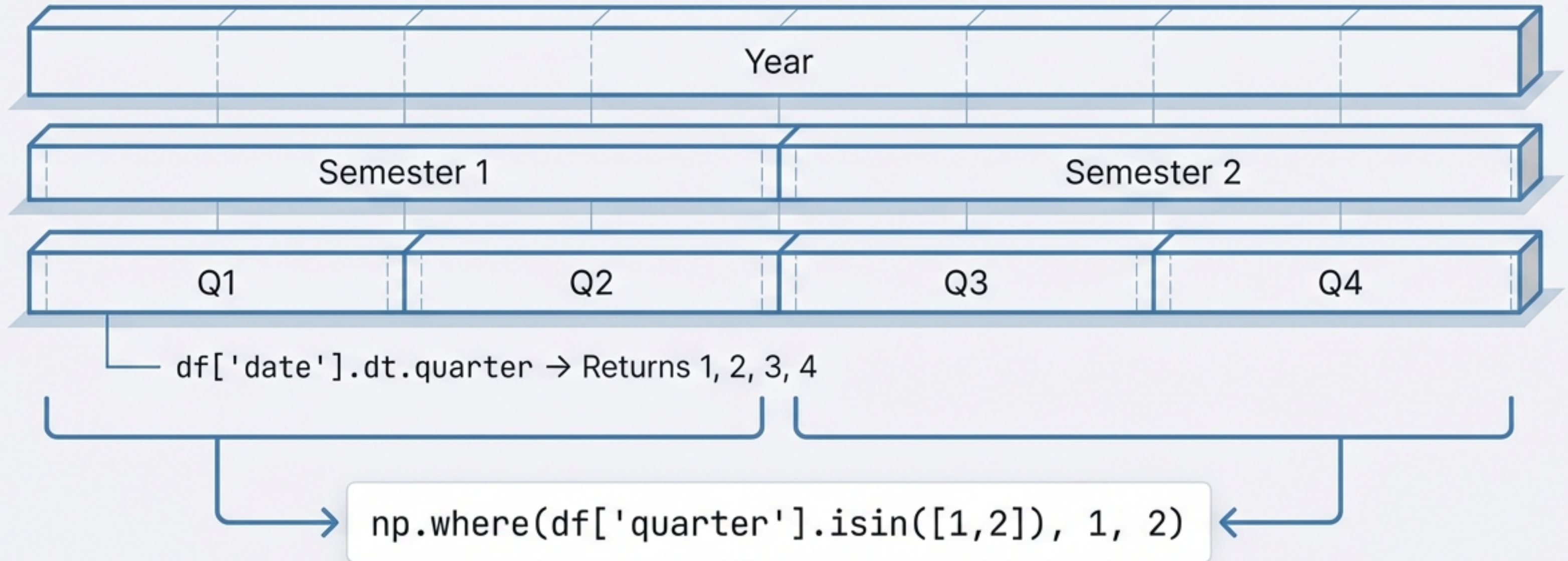


Week of Year

```
df['date'].dt.week
```



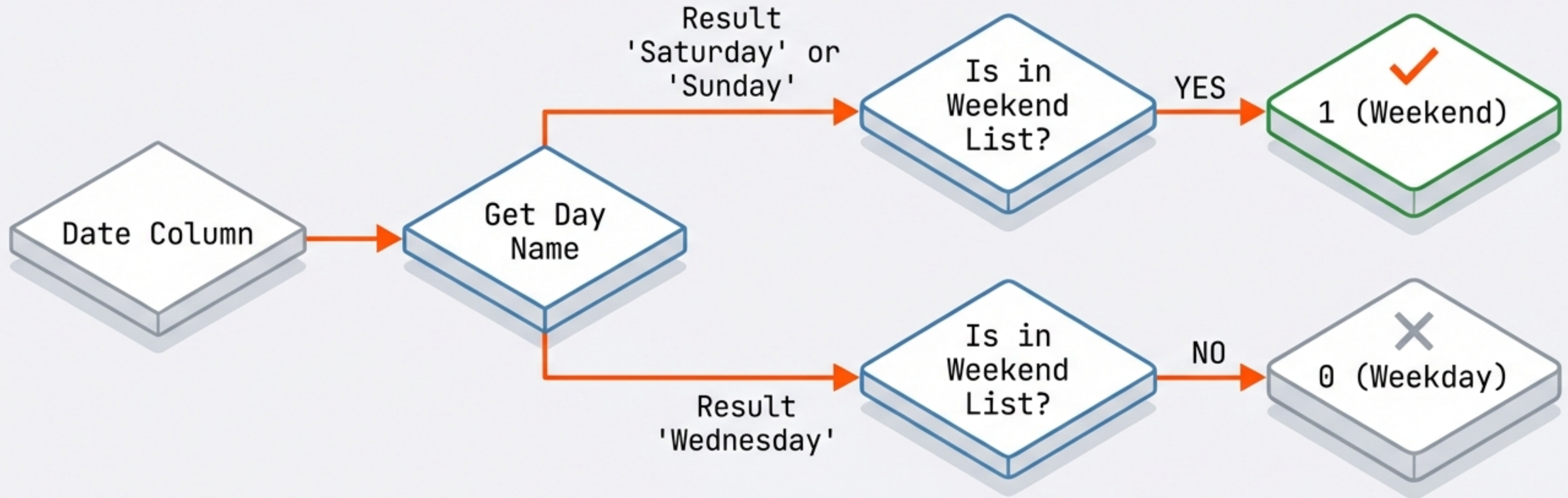
Mapping Dates to Business Cycles



Custom Logic Required (No native function).

Feature Engineering: The Weekend Flag

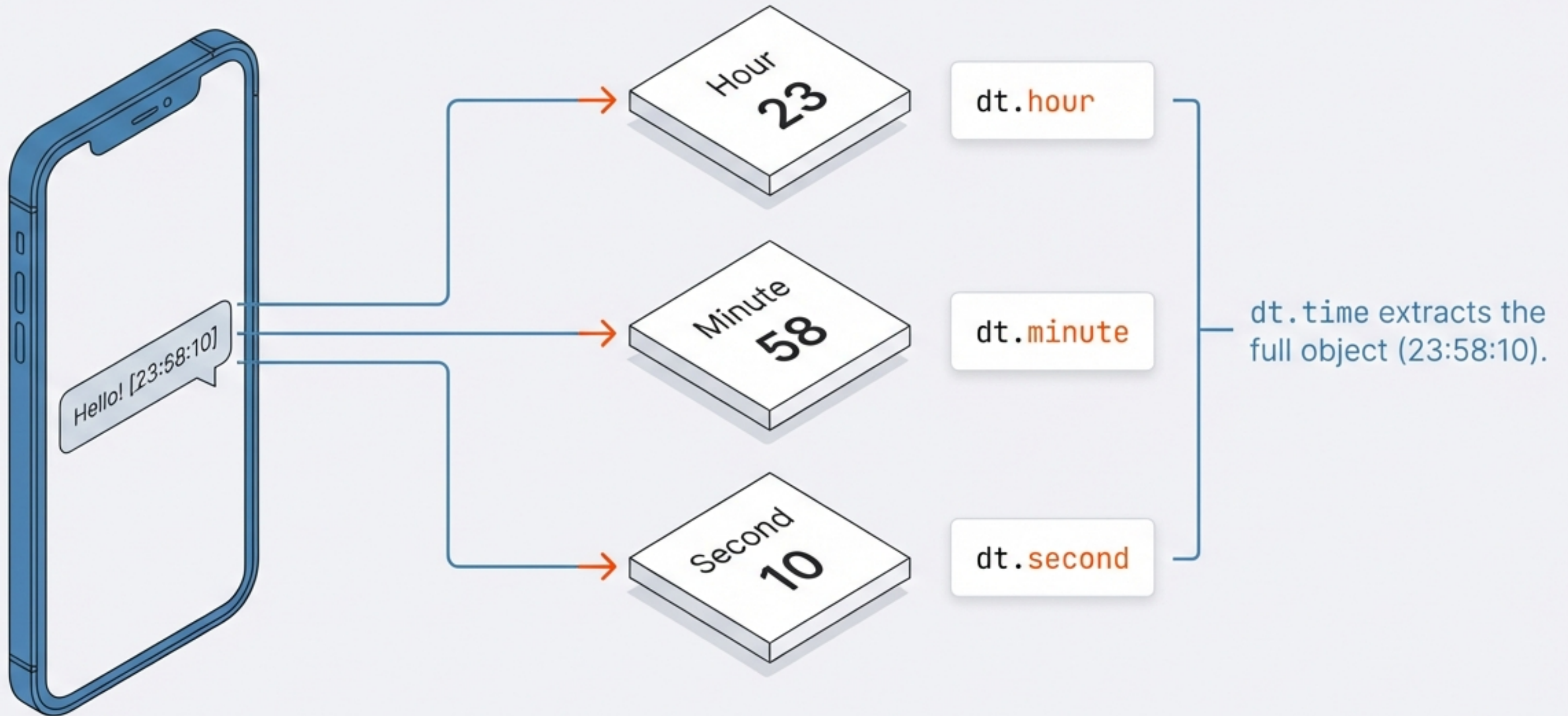
Creating a binary feature for behavioral analysis.



```
df['is_weekend'] = np.where(df['day_name'].isin(['Saturday', 'Sunday']), 1, 0)
```


Granularity: Handling Timestamp Data

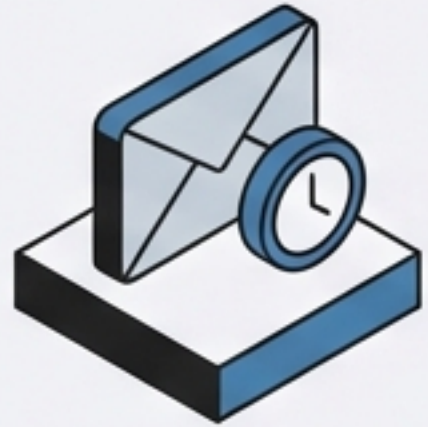
Dataset: messages.csv (High-frequency chat logs)



Time Deltas: Calculating Duration

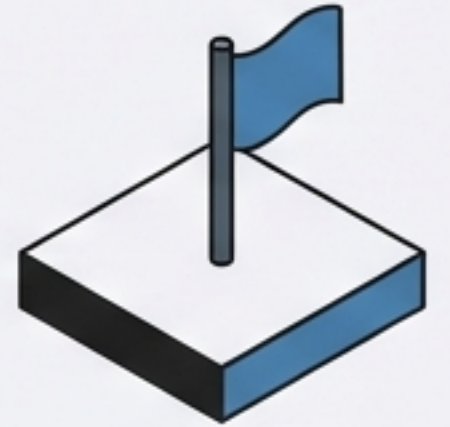
Measuring the time elapsed between an event and today.

$$\text{Delta} = \text{Today} - \text{Event Date}$$



Event Date

JetBrains Mono



Today

JetBrains Mono

```
import datetime
```

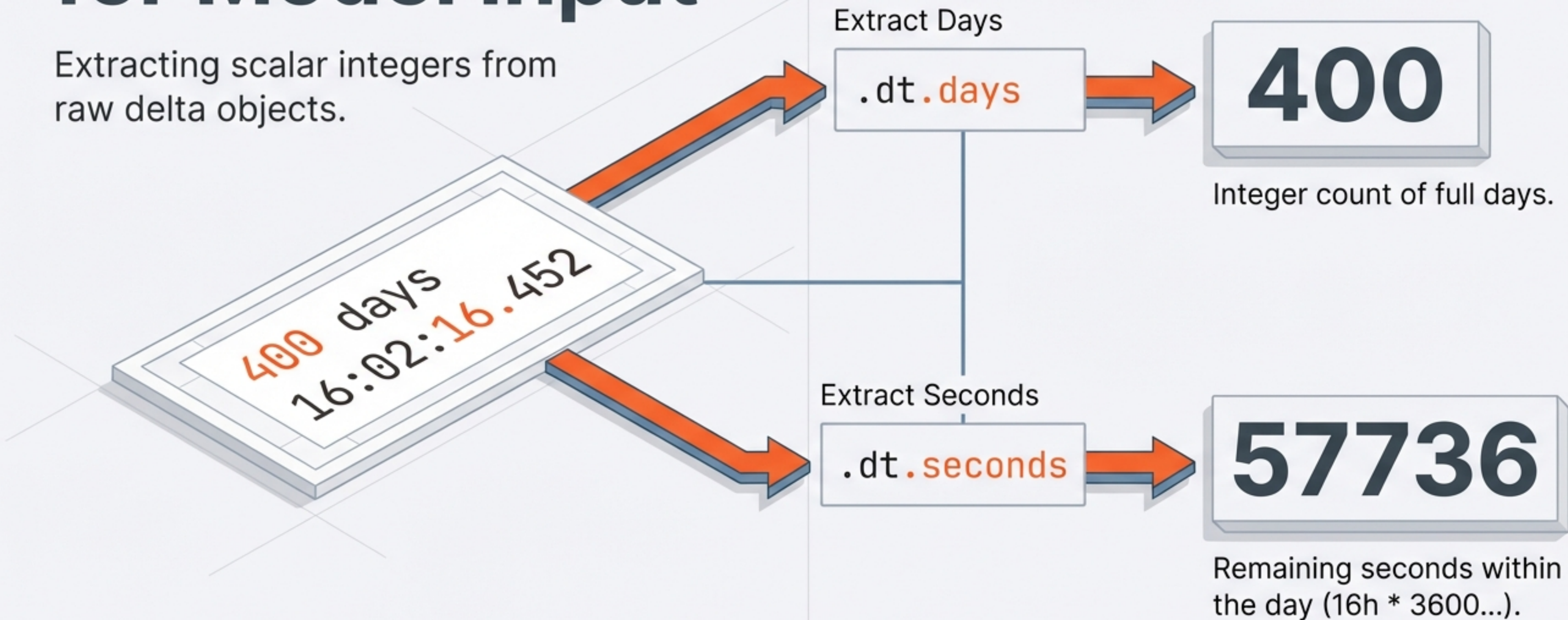
```
today = datetime.datetime.today()  
df['time_elapsed'] = today - df['date']
```

```
0    400 days 16:02:16
```

```
1    380 days 04:12:00
```

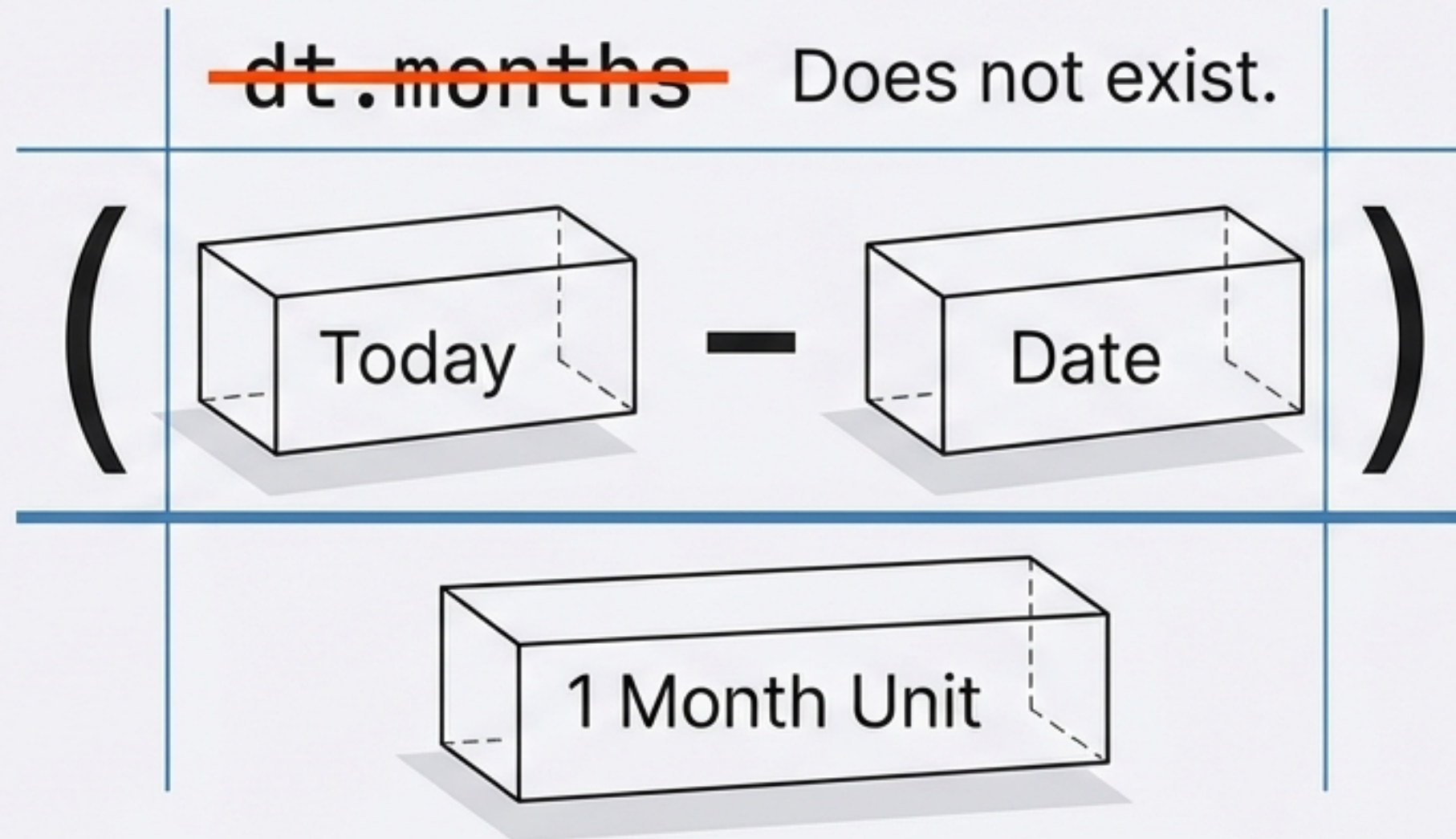

Formatting Deltas for Model Input

Extracting scalar integers from raw delta objects.



Advanced Calculation: Standardizing Units

Using NumPy arithmetic to calculate 'Months Passed'.



Unit Codes:

'M' = Month

'W' = Week

'm' = Minute

```
(today - df['date']) / np.timedelta64(1, 'M')
```


Syntax Reference Guide

Preprocessing

```
pd.to_datetime(df['col'])
```

Basic Components

```
.dt.year  
.dt.month  
.dt.day
```

Names & Strings

```
.dt.month_name()  
.dt.day_name()
```

Calendar Context

```
.dt.dayofweek (0-6)  
.dt.week (1-52)  
.dt.quarter (1-4)
```

Time Components

```
.dt.hour  
.dt.minute  
.dt.second
```

Deltas & Duration

```
(today - df['date'])  
.dt.days  
/ np.timedelta64(1, 'M')
```


The Feature Engineering Workflow



“Hidden information in date columns often holds the key to user behavior—from weekend spending habits to seasonal messaging trends.”